

Exploring LLMs for EEG Classification through Active Prompting with Purity-Guided Unique-Round Example Selection (PURE)

Jingwei Luo, Dingkun Liu, and Dongrui Wu, *Fellow, IEEE*

Abstract—~~With the rapid scaling of large Large~~ language models (LLMs), ~~their have demonstrated impressive~~ in-context learning (ICL) capabilities, ~~particularly especially~~ in few-shot prompting, ~~have attracted considerable attentionscenarios~~. However, ~~existing studies have predominantly most studies have~~ focused on natural language processing (NLP) and computer vision (CV), with limited exploration into complex numerical domains such as ~~Electroencephalography electroencephalogram~~ (EEG) signals. ~~In this papersignal classification. To bridge this gap,~~ we propose a ~~novel method that purity-guided active prompting approach,~~ which integrates domain-specific EEG feature extraction with a ~~purity-guided active selection strategy to identify a small an active learning-based demonstration selection strategy.~~ PURE selects a compact set of prototypical examples ~~from a large sample pool. These examples are embedded into prompts as demonstrations, enabling LLMsto perform EEG classification in a that optimally activate LLMs' few-shot setting without any learning abilities without requiring gradient-based parameter updates. Our approach exhibits three desirable properties. First, to the best of our knowledge, this is the first work to leverage prompt engineering—a simple yet powerful technique—to bridge the gap between LLMs and EEG-based classification, demonstrating its feasibility. Moreover, unlike fine-tuning, prompt engineering remains applicable to both open-source and closed-source LLMs, thereby broadening its practical applicability. Second, while random example selection remains the common practice in most existing studies, we propose an active selection strategy that identifies the most valuable samples for LLMs, achieving significantly better performance and higher stability. Third, unlike conventional NLP tasks where examples must be re-selected for each test instance, our example selection is performed only once at the beginning, substantially reducing computational cost and inference latency.~~ Unlike conventional random selection or per-instance demonstration selection methods, PURE identifies highly representative EEG samples once from the training dataset, significantly reducing computational overhead. Extensive experiments ~~across multiple EEG datasets, BCIparadigms, and LLMs demonstrate that our method consistently outperforms random selection baselines, achieving superior accuracy and stability. Furthermore, performanceimproves with more powerful LLMs (e. g., DeepSeek-V3), and in some cases, our method even surpasses state-of-the-art supervised models, on seven EEG datasets and three brain-computer interface (BCI) paradigms demonstrated that PURE achieved superior performance.~~

Index Terms—EEG, LLMs, active learning, prompt engineering, in-context learning, motor imagery, few-shot prompting,

J. Luo, D. Liu, and D. Wu, are with the Ministry of Education Key Laboratory of Image Processing and Intelligent Control, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China. They are also with the Shenzhen Huazhong University of Science and Technology Research Institute, Shenzhen, 518000 China.

Corresponding Authors: Dongrui Wu (drwu09@gmail.com).

domain knowledge fusion

I. INTRODUCTION

Transformer-based [1] pre-trained large language models (LLMs) have achieved ~~remarkable progress across a wide range of significant advancements across diverse~~ natural language processing (NLP) tasks. As ~~their size increases, these models gain the ability to perform these models scale in size, they exhibit notable~~ in-context learning (ICL), ~~where a small number of capabilities, where few~~ demonstration examples provided at inference time enable the model to achieve ~~state-of-the-art (SOTA) or near-SOTA performance —good performance~~ without any parameter updates [2].

However, in—

However, ~~purely textual descriptions are insufficient for many real-world applications, language alone is insufficient.~~ prompting the need to integrate additional numerical or structured information directly into textual prompts. Different tasks often rely on ~~non-linguistic modalities to convey information, and textual inputs may not fully capture the necessary information~~ diverse types of information beyond plain text. To bridge this gap, various efforts have been made to integrate LLMs with other modalities such as images [3], [4], audio [5], video [6], and physical actions [7] ~~to augment LLM inputs.~~ Nevertheless, applying LLMs to EEG signals presents a distinct set of challenges. Firstly, EEG data suffers from limited high-quality annotations, especially aligned EEG-text pairs, which are essential for effective model supervision or alignment. Secondly, EEG signals are inherently non-stationary—, their statistical properties vary over time and across sessions—, posing difficulties for consistent feature extraction and model generalization. ~~Thirdly, there exists significant Third, substantial inter-subject variability, meaning significantly impacts model generalization, resulting in poor performance when transferring models trained on one group of subjects often perform poorly when transferred to another. Finally, EEG recordings are characterized by a low signal-to-noise ratio due to contamination from muscle activity, eye movements, and environmental artifacts. As a result, these challenges make it particularly difficult to integrate LLMs with EEG-based tasks.~~ These complexities significantly hinder the direct use of EEG signals in standard LLM frameworks.

An alternative strategy ~~is to embed involves embedding~~ numerical signals into textual ~~templates, allowing LLMsto~~

~~process them as prompts. Prior studies have explored such methods for structured data like tables [8] or prompts, enabling their processing by LLMs. Studies have successfully applied this method to structured numerical data, such as tables [8] and medical records [9], where values typically carry semantic meaning and clear semantic meanings which can be easily verbalized. In contrast, EEG signals lack inherent~~ However, EEG fundamentally lack semantic interpretability. For instance, the number “16” in a clinical record may indicate a meaningful age, whereas the same value in EEG data might simply represent an arbitrary signal amplitude at a specific time point, devoid of symbolic meaning.

In addition, EEG data is high-dimensional and temporally dense. A ~~2-second two-second~~ EEG trial sampled at 256 Hz across 8 channels yields 4,096 numerical values. Directly inputting such dense, unstructured data into an LLM consumes ~~excessive tokens and hinders the model’s ability to learn meaningful patterns. As a result is impractical, consuming excessive computational resources and hindering meaningful pattern recognition. Therefore,~~ grounding LLMs in low-semantic, large-scale numerical domains like EEG remains underexplored.

~~To address these challenges, incorporating domain knowledge to extract discriminative and compact features from raw EEG signals is both essential and effective. In our experiments on MI tasks—which primarily elicit event-related desynchronization (ERD) and synchronization (ERS) patterns in the μ (8–13 Hz) and β (13–30 Hz) frequency bands over the sensorimotor cortex, particularly around electrodes C3 and C4—we drew inspiration from the Filter Bank Common Spatial Pattern (FBCSP) framework [10]. Specifically, we found that applying bandpass filtering in the 8–24 Hz range, dividing the spectrum into four 4-Hz sub-bands, and extracting two CSP features from each sub-band significantly enhanced classification performance.~~

~~Just as in NLP tasks, selecting appropriate in-context examples is crucial for model performance [11]. Prior work has proposed selecting examples based on semantic similarity [12] or using clustering to balance similarity and diversity [13]. These methods aim to identify examples that are both informative and representative, maximizing the full potential of LLMs. However, we find that such strategies are not directly effective for EEG signal classification. Based on extensive experimentation and analysis, we propose a new example selection method tailored for EEG features and other types of complex numerical data. Our method differs from standard NLP in-context selection approaches in two key ways.~~

~~First, instead of selecting examples for each test instance individually as commonly done in NLP, we select a fixed set of examples from the training pool once at first, and reuse them for all test samples. This significantly reduces computational overhead. Second, in the context of EEG signal classification, we find that prompting the LLM with class-consistent, low-entropy examples from each category consistently improves accuracy. In contrast, using high-entropy, hard-to-predict samples results in poor performance, while mid-level examples—such as those near the cluster center—fall somewhere in between.~~

~~Based on these observations, we~~

We propose a simple yet effective algorithm that selects high-confidence examples by estimating local same-label purity—defined as the proportion of same-label neighbors within a k -nearest neighborhood. This selection strategy identifies examples with the highest intra-class consistency for prompting, and achieves significantly better and more stable performance than random selection across multiple BCI paradigms and EEG datasets.

In summary, our contributions in this paper are:

- 1) To the best of our knowledge, this work is the first to directly explore the feasibility of using LLMs for classifying EEG signals via in-context few-shot prompting, thereby extending the application of LLMs to complex, low-semantic numerical domains beyond natural language.
- 2) We incorporate domain-domain-specific knowledge to extract compact and task-relevant features from raw features from EEG signals. This not only reduces the number of input tokens substantially but also improves classification performance, making the LLM-based prompting pipeline more effective and computationally feasible.
- 3) Unlike prior work in NLP that performs example selection per test instance, we introduce a fixed example set that is selected once from the training pool and reused across all test samples. This strategy significantly reduces computational cost and inference latency.
- 4) ~~We propose a novel example selection method named PURE, which identifies low-entropy, class-consistent samples based on local same-label purity. Our findings challenge common assumptions regarding criteria used for demonstration selection, and demonstrate that such examples consistently lead to superior and more stable performance across multiple BCI paradigms and EEG datasets.~~ Extensive experiments on seven EEG datasets and three brain-computer interface (BCI) paradigms demonstrated that PURE achieved superior performance.

The remainder of this paper is organized as follows: Section II introduces related work. Section III proposes PURE. Section IV presents the experiment results. Finally, Section VI draws conclusions.

II. RELATED WORK

This section introduces related works on ...

A. Few-shot Prompting

III. RELATED WORKS

A. Few-shot Prompting

With the rapid advancement of LLMs, ICL has emerged as a powerful paradigm that enables models to perform downstream tasks by conditioning on a small set of input-output examples without updating model parameters. A particularly effective and widely adopted form of ICL is few-shot prompting, wherein a handful of labeled demonstrations are provided in the prompt during inference. This approach has achieved

competitive, and sometimes even ~~SOAT-SOTA~~ results across diverse tasks.

A. Demonstration Selection for Few-shot Prompting

Despite its simplicity, the success of few-shot prompting depends heavily on the selection and arrangement of demonstration examples. Poor example quality or ordering can lead to degraded performance or unintended biases.

In existing studies on demonstration selection for ICL, several strategies have been proposed. Liu *et al.* [12] ~~proposed to select demonstrations based on similarity, retrieving the k suggested selecting examples based on semantic similarity, proposing to retrieve the K nearest neighbors of a test instance~~ using sentence encoders to compute distances in the embedding space. Li *et al.* [13] ~~incorporate~~ incorporated diversity by clustering the sample pool and then selecting the nearest sample to the test instance from each cluster. In the field of CV, recent work has extended this idea by assigning a similarity-based score (e.g., cosine distance) to each candidate, reflecting its suitability for inclusion in the prompt [14].

Overall, current research on ICL demonstration selection is primarily focused on NLP and CV tasks, with selection criteria centered around similarity to the query, representativeness, diversity, and uncertainty.

B. EEG-based Classification

EEG-based classification tasks cover a wide range of scenarios, such as motor imagery, sleep stage classification, and epilepsy detection. Traditional approaches typically rely on feature extraction methods like CSP, followed by classifiers such as logistic regression (LR) or support vector machines (SVM). In recent years, deep learning models—particularly those based on convolutional neural networks (CNNs)—have been proposed and have achieved ~~SOAT-SOTA~~ performance across various tasks. However, due to the limited size of available EEG datasets, the scale of these models remains constrained, making it infeasible to train models comparable in size to LLMs or large vision models (LVMs).

Jiang *et al.* [15] introduced a 0.369B-parameter model trained on a large collection of EEG data spanning multiple paradigms and tasks, which represents one of the largest models in the EEG domain to date. Nonetheless, its performance and ease of use in downstream tasks still lag far behind that of LLMs. Due to the modality gap, LLMs cannot be directly applied to EEG-based tasks. To the best of our knowledge, no prior work has directly applied prompt engineering to EEG classification tasks.

III. METHOD

~~In our setting, we start~~

This section introduces our proposed PURE algorithm. Specifically, we begin with a large pool of labeled training samples ~~. Our objective is to select a small subset of these samples and select a compact subset~~ to serve as in-context examples within the prompt, aiming to ~~best activates fully leverage~~ the few-shot capabilities of ~~a LLM and maximizes the~~

LLM and maximize its performance on the test set. Throughout the entire prediction process, ~~we do not update the model's parameters, ensuring the simplicity of the overall workflow. Before introducing our method, we begin by re-examining traditional sample selection approaches from two perspectives: no gradient-based updates are performed, allowing the method to operate in a purely forward-pass manner and ensuring a streamlined and efficient framework.~~

A. ~~Per-Instance Selection Compared to One-Time Selection~~

~~In studies on ICL, particularly those in the fields of NLP and CV [12], [14], most existing works perform example selection separately for each test instance, often using methods such as k NN or clustering to identify similar and representative examples. We refer to this approach as per-instance selection.~~

~~However, in numerical data classification tasks, using a k NN-based per-instance selection strategy can easily bias the LLM toward predicting the majority label among the selected examples. This often causes the model to behave similarly to a standard k NN classifier, thereby undermining the unique advantages of LLMs.~~

~~In fact, in order to demonstrate that the ability of LLMs is not solely driven by k NN, [12] introduces a baseline where k NN alone is used for prediction. The results show that k NN performs significantly worse than k NN+LLM. However, many of the tasks considered involve open-ended output spaces—for example, text-generation tasks. As described in [12], “for text generation tasks, the target y_1 associated with the first retrieved example is directly used as the predicted output.” Therefore, the mismatch between the retrieved target and the true output is almost inevitable, even when the examples are close in the embedding space.~~

~~Thus, the relatively poor performance of k NN in such tasks is expected and does not invalidate the potential overlap between k NN and LLM. If this bias is relatively subtle in non-classification tasks, it becomes especially evident in classification tasks.~~

~~Subsequent experiments further confirm this observation. Although k NN-based per-instance selection can achieve reasonably good prediction accuracy, its performance is often similar to that of a standard k NN classifier. In most cases, unless a very large model is used, it usually performs worse than k NN. Moreover, the cost of per-instance selection is substantial. It requires actively selecting examples for each test sample individually, often involving multiple iterations per selection. This results in high computational and time costs.~~

~~To address these issues, we propose a one-time selection approach. Instead of selecting examples repeatedly for each test instance, we select a fixed set of examples once from the training set at the beginning and use this same set for all subsequent predictions. This method offers significant advantages over per-instance selection in terms of efficiency and scalability, particularly in the following two aspects:-~~

- 1) ~~Reduced computational overhead.~~ It eliminates the need for repeated per-instance selection, avoiding the iterative process for each test sample.

- 2) *Improved batch efficiency.* Since all test samples share the same in-context examples, we can batch multiple test inputs into a single prompt and obtain multiple predictions with one API call. In contrast, per-instance selection requires constructing one prompt per test instance, yielding only a single prediction at a time.

Furthermore, in terms of prediction accuracy, our experiments show that performance improves as the size and capability of the LLM increase. On some datasets, the one-time selection strategy even outperforms SOAT methods.

A. Gold Criteria for Demonstrations

Once the one-time selection strategy is adopted, the notion of similarity—which has served as a primary criterion in most prior few-shot demonstration selection methods—no longer applies, as there is no specific test instance to serve as a reference for measuring similarity. This calls for a major shift: we must focus on samples that best represent the entire input space rather than resemble an individual instance.

In conventional classification frameworks, constructing a robust decision boundary is critical for achieving strong generalization performance. Accordingly, three core criteria are widely adopted in active learning:

- 1) *Informativeness.* Informativeness quantifies how challenging a sample is for the model to classify, typically indicating that it carries substantial information. Informative samples are expected to exert a notable influence on the refinement of the decision boundary. This criterion can be measured using various approaches, such as model uncertainty (e.g., predictive entropy, distance to the decision boundary, or confidence scores), expected model change, or expected error reduction.
- 2) *Representativeness.* Representativeness measures how well a sample reflects the overall distribution of the input space, often based on its similarity to other samples [16]. Representative samples are typically not outliers. A common strategy is to perform k -means clustering over the unlabeled pool and select samples closest to cluster centroids as proxies for the underlying data structure. In regression tasks, representativeness aids in capturing input space structure; however, for classification tasks, it plays a secondary role to informativeness in defining the decision boundary.
- 3) *Diversity.* Diversity ensures that selected samples span a wide range of the input space, rather than clustering in a narrow region. In classification, this generally involves selecting samples along the decision boundary to maximize coverage. Techniques such as greedy sampling and k -means clustering are frequently employed to enhance sample diversity.

Results of random, informativeness and representativeness Method accuracy random 0.541 informativeness 0.343 representativeness 0.583

However, when applying LLMs to EEG-based classification tasks, it remains unclear whether these principles still apply. To address this, we conducted a series of experiments to examine

how effectively various selection strategies can activate the few-shot learning capabilities of LLMs.

Concretely, we evaluate three selection methods: (1) random selection as a baseline, (2) entropy-based selection, which identifies the most informative examples based on prediction uncertainty, and (3) k -means clustering, which selects examples that are both representative and diverse. For all methods involving distance computations, the Euclidean distance is used. Experiments are conducted on the well-established BCI Competition IV 2a dataset, focusing on a binary-class motor imagery task. The training and test sets each contain 144 trials, drawn from two separate sessions, with balanced class distributions.

Additionally, since only a small number of examples are used in few-shot prompting, we adopt a class-balanced selection strategy in all experiments to prevent the selected examples from being dominated by a single class, which could bias the LLM’s predictions. All methods are repeated 25 times, and the results are reported in Table ??.

As shown in Table ??, using k -means-selected samples as in-context examples leads to overall better prediction performance than random selection. This indicates that strategies based on representativeness and diversity are also effective for in-context example selection in LLMs. However, contrary to our expectations, using the most informative samples—those with the highest entropy—as in-context examples yields surprisingly poor results, with an average accuracy of only 38.89%, significantly lower than that of random selection. This suggests that information-based selection strategies, which are among the most effective in traditional classification models, may completely fail in the context of in-context learning with LLMs.

A t-SNE visualization of the training samples from subject 8 of the BNCI2014001 dataset, with different colors representing two distinct classes. Red squares indicate the samples selected through point-by-point evaluation with the LLM, which achieved the best prediction results when used as demonstrations.

To better understand which types of samples contribute most to model performance, an additional set of experiments was conducted. Each sample in the training set was individually used as the sole in-context example, and the corresponding prediction accuracy was recorded. Based on these results, all samples were ranked by performance, and the best-performing ones were selected. Their distribution was then analyzed using t-SNE, as shown in Figure ?? (Subject 8). Interestingly, these high-performing samples are not located near the decision boundary; on the contrary, most lie far from it. This pattern was consistently observed across multiple subjects.

Motivated by this observation, we took the opposite approach by selecting the least informative samples—those predicted by the model with the highest confidence. Using the same evaluation procedure, the results are reported in Table reftab:pre-exp. Remarkably, these low-entropy samples not only outperform random selection but also achieve better performance than the k -means-based selection strategy. Furthermore, to mitigate the risk of selecting outlier samples, we leveraged the known label information and proposed a

local same-label purity method. This strategy further enhances the performance of the LLM, as demonstrated in subsequent experiments.

We hypothesize that, for classification tasks involving numerical data, few-shot prompting in LLMs relies more on analyzing the intrinsic characteristics of the data itself, rather than constructing a deep neural network as in fine-tuning. Consequently, more prototypical and easily classifiable examples may better represent the underlying characteristics of each class, thereby enabling more accurate predictions. In contrast, highly informative examples—those near the decision boundary and more difficult to classify—may introduce ambiguity and offer less effective guidance during inference.

Flowchart of EEG-based classification using LLM few-shot prompting.

A. Purity-based Unique-Round Example Selection (PURE)

Based on the preceding observations, we propose PURE (Purity-based Unique-Round Example Selection), a cost-effective strategy for selecting in-context examples in EEG-based tasks. The overall procedure is illustrated in Figure ??.

We begin by applying standard preprocessing to the raw EEG signals, followed by task-specific hand-crafted feature extraction informed by domain knowledge. Specifically, for MI tasks, the EEG signals are band-pass filtered into 4 sub-bands within the 8–24 Hz range, each with a 4 Hz bandwidth. The CSP algorithm is then applied to each sub-band to extract 2 features, resulting in an 8-dimensional feature vector for each trial.

Next, for each feature vector in the training set, we identify its k nearest neighbors and compute the proportion of neighbors that share the same class label—this value is referred to as the local same-label purity. The pseudocode of PURE is given in Algorithm 1. All training samples are then ranked based on their purity scores, and the top- n most “pure” samples are selected as in-context examples. This purity-based selection strategy not only identifies the most prototypical samples within each class but also helps to avoid outliers, as samples surrounded by neighbors from different classes tend to receive lower purity scores and are thus less likely to be selected.

Moreover, although the method does not explicitly enforce diversity, we empirically observe that augmenting it with greedy diversity sampling degrades performance, suggesting that purity alone is a more effective criterion in this context.

Finally, the selected examples, along with their corresponding labels, are combined with task descriptions, background context, and explicit output instructions to construct the full in-context prompt, which is then provided to the LLM alongside the test input.

IV. DATASETS

This section introduces five MI datasets, one sleep dataset, and one epilepsy dataset used in our experiments.

Algorithm 1: Purity-based Unique-Round Example Selection (PURE)

Input: N labeled samples, $\{\mathbf{x}_n\}_{n=1}^N$, where $\mathbf{x}_n \in \mathbf{R}^d$;
 C , number of classes;
 M , number of samples to select;
 K , neighborhood size

Output: A list of selected sample indices with balanced class distribution L .

Ensure M is divisible by C ;

foreach sample $n = 1$ to N **do**

 Compute distances between $\mathbf{x}_n$ and all other samples;

 Find K nearest neighbors of $\mathbf{x}_n$ (excluding itself);

 Compute same-label purity score:

$$\text{purity}(\mathbf{x}_n) = \frac{\text{Number of same-label neighbors}}{K}$$

end

Group all samples by class label;

Sort each group in descending order of purity score;

foreach class $c = 1$ to C **do**

 Select top M/C samples with highest purity;

 Add their indices to output list L ;

end

return Selected sample indices list L .

A. MI Datasets

All MI datasets used in our experiments are binary classification tasks and were obtained using the MOABB framework [17]. Specifically, two datasets from BCI Competition IV (BNCI2014001 and BNCI2014004) [18] contain left-vs-right hand imagery, while the other three—BNCI2014002, BNCI2015001, and Weibo2014—contain right-hand vs. feet imagery.

Their experimental paradigms were similar: In each session, a subject sat comfortably in an armchair positioned in front of a computer screen. At the beginning of each trial, a fixation cross appeared on a black background, accompanied by a short auditory warning tone. After a brief delay, a directional cue in the form of an arrow pointing left, right, down, or up appeared on the screen, corresponding to one of the four MI classes: left hand, right hand, feet, or tongue. This cue remained visible throughout the task period and served as a prompt for the subject to begin the corresponding MI task. Subject performed the imagery task until the fixation cross disappeared. Each trial ended with a brief black screen before the next trial began [18].

- 1) *Left Hand vs. Right Hand Datasets.* Dataset BNCI2014001 comprises recordings from 9 subjects, with EEG signals collected from 22 channels at a sampling rate of 250 Hz. Although the original paradigm involves four MI tasks (left hand, right hand, feet, and tongue), only the left- and right-hand trials are used in our experiments. Each subject completed two sessions on different days, with 72 trials per class per session, totaling 144 trials per session. Dataset BNCI2014004 also includes 9 subjects but uses

TABLE I
SUMMARY OF THE SEVEN EEG DATASETS.

BCI Paradigm	Dataset	Number of Subjects	Number of Channels	Sampling Rate (Hz)	Trial Length (seconds)	Number of Trials per Subject	Class Labels
MI	BNCI2014001	9	22	250	4	288	left hand, right hand
	BNCI2014004	9	3	250	4.5	680-760	left hand, right hand
	BNCI2014002	14	15	512	5	160	right hand, both feet
	BNCI2015001	12(9)	13	512	5	400/600	right hand, both feet
	Weibo2014	10	60(22)	200	4	140, 160	right hand, both feet
Sleep	Sleep-EDFx	9	2	100	30	1025	W, N1, N2, N3, REM
Epilepsy	TUEP	10	23	250	30	390	epilepsy, non-epilepsy

only 3 bipolar EEG channels (C3, Cz, and C4), sampled at 250 Hz. Each subject participated in 5 sessions: the first 2 (screening) without feedback, and the last 3 with real-time feedback. During the feedback sessions, a smiley face was displayed and changed color based on the accuracy of the subject’s imagined movement.

- 2) *Right Hand vs. Feet Datasets*. BNCI2014002 includes 14 participants and follows a similar paradigm to BNCI2014001. Each subject completed 1 session with 8 runs (5 for training and 3 for evaluation), totaling 160 trials.

BNCI2015001 adopts the Graz two-class synchronous MI paradigm, using EEG recorded from 3 Laplacian derivations. Data were sampled at 512 Hz and filtered using a 0.5–100 Hz bandpass filter with a 50 Hz notch filter. The last 3 subjects were excluded due to poor performance.

Weibo2014 contains data from 10 right-handed participants who underwent a one week training phase prior to EEG recording. EEG was collected from 64 channels placed according to the international 10–20 system, bandpass filtered between 0.5–50 Hz, and downsampled to 200 Hz. Each participant completed 9 blocks of trials involving 6 MI tasks. Only right-hand and feet trials were used for binary classification in this study. From the original 60 usable channels, we selected 21 MI-related channels: FC3, FC1, FCz, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P3, P1, P2, and P4.

B. Sleep Datasets

The Sleep-EDF Expanded (Sleep-EDFx) dataset is intended for sleep research, particularly for sleep stage classification. It contains 197 full-night polysomnographic (PSG) recordings, including EEG (Fpz–Cz and Pz–Oz), EOG, EMG, and event markers. Corresponding hypnograms (sleep stage annotations) were manually scored by trained technicians following the Rechtschaffen and Kales manual and are provided alongside the recordings.

The data originate from two studies: the Sleep Cassette Study and the Sleep Telemetry Study. The main difference is that the Sleep Cassette Study investigates the effect of age on sleep in a home environment over 2 nights (approximately 20 hours each), while the Sleep Telemetry Study examines the effect of temazepam (a sleep medication) in a hospital

setting over 2 nights (approximately 9 hours each), using a high-quality telemetry system.

In our experiments, we focused on data from the Sleep Cassette Study and used only EEG channels for a five-stage sleep classification task. Due to the large volume of data, we selected recordings from 9 subjects, using 1 night of data per subject.

During the signal processing and feature extraction stages, EEG data from the Sleep-EDFx dataset were first bandpass filtered between 0.5 Hz and 35 Hz. Since the recordings span an entire night and include prolonged wake periods at the beginning and end, only the first and last 30 minutes of wakefulness were retained to reduce redundancy. Sleep stages were labeled according to the AASM standard and grouped into five classes: Wake (W), N1, N2, N3, and REM. Each trial was segmented into non-overlapping 30-second epochs.

To extract frequency-domain features, four bandpass filters were applied to isolate the following frequency bands: 0.5–4 Hz, 4–8 Hz, 8–12 Hz, and 12–16 Hz. For each band, the root mean square (RMS) power was computed on a per-channel basis, defined as the mean of the squared signal amplitude across all samples within the trial. This process yielded a feature representation with shape $(N, C \times B)$, where N is the number of trials, C is the number of EEG channels, and $B=4$ is the number of frequency bands.

C. Epilepsy Datasets

The TUH EEG Epilepsy Corpus (TUEP) dataset is a subset of the TUEG collection and includes 100 subjects diagnosed with epilepsy and 100 non-epileptic subjects, as determined by certified neurologists. Since our task involves classifying between epileptic and non-epileptic cases, the experiments were conducted in a cross-subject setting. Specifically, we selected 5 subjects from each group and used 1 session per subject to construct the experimental dataset.

V. EXPERIMENTS

Extensive experiments are conducted in this section to validate the effectiveness of PURE. We use the Qwen-2.5-14B-Instruct-1M model as the LLM, and apply two demonstrations per category, resulting in a total of four ICL demonstrations for the MI binary classification dataset, ten for the sleep five-class classification task, and four for the epilepsy binary classification dataset.

A. Comparing Method

- 1) *BL*, which randomly selects all M samples.
- 2) *CTD*, which applies the k -means algorithm within each class and selects centroids.
- 3) *RD* [19], which starts by selecting the centroid of each class as the initial point. Then, iteratively performs additional rounds of clustering on each class and selects the centroid of the largest cluster not yet covered by the previously selected points.
- 4) *SVM-near*, which trains a SVM classifier on the training set and selects, from each class, the sample closest to the decision boundary.
- 5) *SVM-far*, which in contrast to the above method, selects the sample farthest to the decision boundary within each class. To mitigate the impact of outliers, the method further filters candidates by checking the proportion of same-label samples among their five nearest neighbors.
- 6) *PURE*, which selects the highest-purity samples from each class as describe in section III-A

B. Experimental Setup

To comprehensively evaluate our method, we conducted experiments under three different settings. For the MI datasets, we adopted (i) within-subject, cross-session evaluation and (ii) cross-subject evaluation. For the sleep dataset, as only one session per subject is available, we employed (iii) within-subject five-fold cross-validation instead of (i). For the epilepsy dataset, only the cross-subject evaluation was performed, as each subject contains either epilepsy or non-epilepsy EEG signals.

In the first two settings, each method was repeated five times to ensure stability. Due to the high variability introduced by random selection, experiments involving random baselines were repeated twenty times. For five-fold cross-validation experiments, the entire five-fold process was repeated five times, resulting in a total of 25 runs per method. The average classification accuracy was used as the performance metric.

- 1) *Within-Subject Cross-Session*: In this setting, we used the training and testing splits predefined by each dataset. This approach avoids session leakage while ensuring realistic session-wise generalization. However, the ratio between training and testing samples varies across datasets. Specifically:
 - BNCI2014001: One session is used for training and another for testing, each containing 144 trials.
 - BNCI2014004: Five sessions are available, with the first three used for training and the last two for testing.
 - BNCI2014002: A single session with eight runs; the first five runs are used for training and the remaining three for testing. Each run contains 10 trials for right-hand (RH) and 10 for feet imagery.
 - BNCI2015001: Most subjects have one training session and one testing session, totaling 400 trials. Subjects 8 and 9 have three sessions, where the first two are used for training and the last one for testing.

- Weibo2014: This dataset contains only one session, but since the distribution and quantity of both classes are balanced between the first and second halves, the first half is used for training and the second half for testing.

- 2) *Within-Subject five-Fold Cross-Validation*: For this setting, we employed the `StratifiedKFold` utility from Scikit-learn. All trials of each subject were shuffled and split into five folds, ensuring that the class proportions in each fold matched those of the full dataset. This approach also maintains a consistent ratio of training to testing samples across all datasets.
- 3) *Cross-Subject*: A leave-one-subject-out five-fold cross-validation strategy was adopted. In each fold, one subject was reserved for testing, and the remaining subjects were combined as the training set. To mitigate inter-subject variability in EEG distributions, Euclidean Alignment (EA) [20] was applied to the data from each subject.

C. Results Analysis

Tables II, III, IV, V, and VI presents the within-subject and cross-subject classification accuracies on the five MI datasets, respectively. We can observe that:

- 1) Active selection of ICL demonstrations significantly outperforms random selection. In both within-subject and cross-subject settings, actively selected demonstrations consistently lead to a better prompting effect on the LLM, resulting in higher prediction accuracy and more stable outputs compared to random selection.
- 2) High-informativeness samples yield the worst prompting effect in ICL. Contrary to conventional expectations in traditional machine learning, selecting the most informative samples leads to an inferior prompting effect on the LLM, even worse than random selection. This suggests a fundamental difference between how LLMs and traditional classifiers benefit from informative instances.
- 3) Low-informativeness samples achieve the best prompting effect. Demonstrations with low informativeness and high model confidence, as exemplified by SVM-far and PURE, result in superior prompting effects. We hypothesize that when classifying complex numerical data, LLMs primarily rely on statistical distributional patterns rather than constructing deep decision boundaries. Therefore, more prototypical and easily distinguishable examples are more effective in guiding the model, whereas highly ambiguous samples fail to provide meaningful distinctions.
- 4) PURE consistently outperforms SVM-far. Although SVM-far has been optimized to avoid outlier selection and shows competitive performance, PURE still achieves better overall prompting effects across datasets and experimental conditions.
- 5) PURE achieves consistent superiority across multiple datasets and paradigms. Our proposed method PURE consistently outperforms random selection across all experiments and achieves optimal results in the majority of cases. Specifically, in the within-subject set-

TABLE II

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014001. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	49.31	49.03	67.64	56.11	48.19	55.28	52.22	56.81	83.47	57.56
RD	55.42	48.89	77.92	64.44	48.06	53.19	54.31	71.11	80.14	61.50
CTD	56.81	44.17	71.53	62.92	50.28	51.81	54.58	76.25	86.11	61.61
SVM-near	38.89	50.00	45.00	53.06	50.56	47.36	48.19	46.11	71.53	50.08
SVM-far	58.89	47.64	75.00	61.11	52.78	55.69	57.22	73.06	85.00	62.93
PURE	67.08	51.94	68.61	68.75	52.50	59.44	58.19	77.22	86.67	65.60 (↑8.04)
Cross-subject classification										
BL	48.68	50.42	66.25	59.93	53.33	55.83	45.69	53.68	38.54	52.48
RD	61.25	51.18	74.44	64.65	57.50	61.81	67.64	70.69	52.85	62.45
CTD	58.13	50.14	63.54	63.82	56.53	54.03	63.47	74.44	76.53	62.29
SVM-near	37.15	47.50	52.99	59.65	48.82	43.26	36.11	42.43	26.67	43.84
SVM-far	60.76	47.85	66.67	61.18	54.37	61.53	58.75	72.08	74.10	61.92
PURE	60.76	52.15	70.07	60.76	54.79	62.92	62.64	68.89	70.83	62.65 (↑10.17)

TABLE III

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014002. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification															
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Avg.
BL	61.33	75.00	72.67	60.67	59.33	68.33	84.00	64.00	88.00	54.67	47.67	57.00	47.33	52.67	63.76
RD	65.00	74.67	76.67	68.00	60.67	84.00	89.67	71.67	88.67	56.33	73.67	47.33	49.00	57.67	68.79
CTD	66.67	78.67	80.33	61.33	66.00	83.00	90.00	51.67	94.33	56.00	67.00	54.33	45.33	54.00	67.76
SVM-near	37.33	47.67	20.67	47.67	41.00	89.00	59.67	60.00	72.33	53.33	81.33	61.67	44.33	56.67	55.19
SVM-far	71.00	69.33	77.33	75.33	64.67	82.33	86.67	70.00	92.33	54.00	81.00	61.33	50.33	52.33	70.57
PURE	63.33	74.67	88.33	77.67	61.67	83.00	88.67	71.67	90.00	55.00	79.00	70.00	46.33	58.00	71.95 (↑8.19)
Cross-subject classification															
BL	55.50	60.50	60.62	56.75	55.88	62.38	60.63	62.63	80.25	61.88	59.50	55.75	56.13	48.75	59.80
RD	56.63	69.00	86.75	61.37	67.50	71.88	85.50	70.37	61.25	64.50	61.75	65.00	57.25	44.75	65.96
CTD	55.63	68.75	85.88	63.00	66.63	73.00	85.25	67.00	82.25	65.25	62.25	67.62	55.50	48.88	67.64 (↑7.84)
SVM-near	56.50	29.38	26.75	45.88	40.00	67.37	32.00	66.87	36.75	45.50	63.87	38.12	44.12	53.25	46.17
SVM-far	56.50	64.50	81.75	57.13	59.12	67.87	81.75	61.50	83.63	56.37	62.50	60.88	53.75	53.38	64.33
PURE	57.38	66.25	86.75	55.13	62.75	69.12	85.00	62.38	85.13	60.75	65.50	66.25	51.38	51.88	66.12

ting, PURE improves over random selection by 8.04%, 8.19%, 6.03%, 3.43%, and 4.09% across the five MI datasets, by 5.44% on the sleep dataset, and by 0% on the epilepsy dataset. In the cross-subject setting, PURE achieves similar improvements: 10.17%, 6.32%, 5.35%, 1.37%, and 1.93% across the five MI datasets, 5.67% on the sleep dataset, and 0% on the epilepsy dataset.

- 6) Representativeness- and diversity-based methods (RD and CTD) also outperform random selection Although RD and CTD perform slightly worse than PURE overall, they consistently outperform random selection across all experiments. Notably, in cross-subject settings, their performance significantly improves, achieving results comparable to or even exceeding PURE on several MI datasets. Moreover, since these methods do not strictly rely on label information (labels are only used to maintain class balance), they exhibit broader applicability to scenarios where label information is limited or unavailable.

D. Impact of the Number of Demonstration Samples

In the preceding experiments, we selected two samples per category as demonstrations. In this subsection, we investigate the impact of varying the number of demonstration samples on performance. The experiments are conducted on the BNCI2014001 dataset. Figure 1 presents the performance trends as the number of demonstration samples increases, from which we observe the following:

- 1) The performance of the LLM improves as the number of demonstration samples increases.
- 2) As the number of demonstration samples increases, the marginal performance gains progressively decrease, and the results across different selection methods tend to converge.
- 3) The results of random selection exhibit large fluctuations across repeated trials, indicating significant instability. This further reflects the sensitivity of LLMs to the choice of demonstrations.
- 4) PURE consistently outperforms random and other selection methods across different numbers of demonstra-

TABLE IV

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014004. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	51.44	55.00	51.00	65.31	54.94	47.62	50.44	58.94	51.31	54.00
RD	55.44	49.57	55.40	75.25	54.00	62.62	52.81	71.50	52.81	58.82
CTD	49.19	56.93	52.87	65.81	50.00	52.19	54.88	56.94	54.31	54.79
SVM-near	49.31	56.93	51.40	47.25	51.25	45.94	42.87	52.75	42.81	48.95
SVM-far	53.25	56.29	54.00	77.25	55.50	54.69	49.75	75.94	57.06	59.30
PURE	51.67	50.14	53.67	76.88	61.44	57.25	52.87	78.25	58.13	60.03 (↑6.03)
Cross-subject classification										
BL	50.92	49.79	50.56	54.73	51.84	49.83	51.53	52.13	54.44	51.75
RD	54.89	52.35	50.44	64.81	50.19	57.50	50.14	56.76	48.17	53.92
CTD	47.00	48.26	50.11	45.08	46.73	51.61	49.72	46.76	52.56	48.65
SVM-near	45.36	48.85	49.14	41.84	47.35	48.33	48.17	59.89	48.67	48.62
SVM-far	59.11	52.88	50.00	68.27	53.14	55.19	52.00	61.47	52.36	56.05
PURE	57.00	52.56	52.64	69.05	55.51	57.14	53.25	64.76	52.03	57.10 (↑5.35)

TABLE V

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2015001. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	74.30	88.30	70.50	82.10	65.00	51.30	76.80	55.00	58.40	69.08
RD	79.60	89.00	55.90	77.40	70.40	54.90	76.50	58.30	54.30	68.48
CTD	79.00	89.60	55.50	78.50	69.30	56.00	74.50	58.30	50.50	67.91
SVM-far	67.00	86.30	83.00	82.60	67.70	52.20	81.70	64.90	62.40	<u>71.98</u>
SVM-near	66.70	60.80	53.00	28.60	40.80	50.40	41.20	45.80	51.60	48.77
PURE	71.80	88.80	70.80	82.80	74.40	51.20	76.00	69.20	67.60	72.51 (↑3.43)
Cross-subject classification										
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	70.25	80.15	68.45	65.55	61.85	56.40	55.35	56.60	64.03	64.29
RD	78.50	83.55	65.85	55.60	68.75	56.65	62.00	56.63	63.27	<u>65.64</u>
CTD	71.90	79.35	66.85	62.70	68.35	55.75	58.50	57.87	63.77	65.00
SVM-far	76.15	76.80	66.00	66.20	62.70	50.15	56.70	56.67	58.67	63.34
SVM-near	73.95	69.45	55.25	54.40	69.00	57.45	57.10	56.77	60.03	61.49
PURE	76.70	85.30	66.15	66.45	63.50	56.25	60.50	53.70	62.40	65.66 (↑1.37)

tions, and exhibits the smallest variance across repeated experiments, indicating the highest stability.

E. Impact of Different Models

To further assess the generalizability of the proposed method, we evaluated its performance on alternative LLMs. Specifically, DeepSeek-V3 was selected for comparative analysis. Experiments were conducted on the BNCI2015-002 dataset, and the corresponding results are presented in Table VIII. The following observations can be made:

- 1) Active selection methods consistently outperform random selection across different LLMs.
- 2) The relative performance hierarchy among all methods remains consistent with previous experiments, confirming the robustness of active selection strategies across different model architectures.
- 3) Employing larger and more powerful models leads to a consistent improvement in accuracy for all methods. Remarkably, on certain datasets, the results surpass the SOAT performance achieved by conventional deep neural networks, highlighting the considerable potential of LLM-based ICL few-shot learning approaches in non-text domains such as EEG signal analysis.

VI. CONCLUSIONS AND FUTURE RESEARCH

TABLE VI

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON WEIBO2014. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification											
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Average
BL	67.50	47.50	70.25	53.00	66.50	68.29	56.25	88.50	84.50	64.50	66.68
RD	67.75	46.50	73.00	59.25	76.50	68.57	63.75	84.00	84.75	71.75	69.58
CTD	74.75	49.50	71.50	54.00	69.75	69.71	55.75	84.00	83.50	67.25	67.97
SVM-far	76.25	59.00	73.50	48.50	67.50	75.43	61.50	77.75	90.00	69.25	<u>69.87</u>
SVM-near	37.75	45.75	60.25	46.25	44.25	49.14	43.50	43.75	88.75	63.00	52.24
PURE	69.50	61.75	76.00	67.50	69.25	71.71	57.25	80.00	87.50	67.25	70.77 (↑4.09)

Cross-subject classification											
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Average
BL	63.69	54.06	62.32	52.19	62.94	63.07	73.07	78.25	78.25	61.50	64.93
RD	71.13	52.37	66.87	51.38	64.50	64.14	75.75	80.37	87.75	64.87	67.91 (↑2.98)
CTD	72.94	48.75	66.75	51.88	62.25	55.86	77.62	63.75	85.75	66.25	65.18
SVM-far	68.37	54.13	72.12	57.87	64.12	63.29	71.12	80.50	83.37	60.38	<u>67.53</u>
SVM-near	63.00	53.75	35.88	46.88	54.87	68.71	58.25	43.00	26.37	60.38	51.11
PURE	69.75	55.13	71.37	55.13	60.50	60.00	73.50	74.25	89.38	59.50	66.86

TABLE VII

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON SLEEP-EDF. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification									
Methods	S1	S2	S3	S4	S5	S6	S7	Average	
BL	29.81	30.93	31.93	28.26	35.18	30.18	31.87	31.17	
RD	26.31	28.76	36.58	29.13	34.81	32.76	35.06	31.92	
CTD	23.92	27.08	34.76	26.28	35.66	29.38	28.30	29.34	
SVM-far	37.83	28.43	36.49	31.51	39.63	39.25	37.95	<u>35.87</u>	
SVM-near	28.65	25.85	27.47	26.07	31.34	31.32	26.40	28.16	
PURE	35.57	33.93	35.71	35.97	50.78	31.48	32.80	36.61 (↑5.44)	

Cross-subject classification									
BL	23.59	21.81	19.52	24.87	27.20	20.36	23.18	22.93	
RD	22.67	24.41	17.25	18.27	22.47	18.01	22.15	20.89	
CTD	21.49	20.47	19.62	18.79	19.52	18.10	25.96	20.85	
SVM-far	24.81	23.24	17.98	19.89	22.68	16.16	32.36	22.73	
SVM-near	20.44	21.56	16.13	27.30	22.83	30.13	23.01	<u>23.06</u>	
PURE	29.79	25.81	26.28	20.37	34.64	34.66	24.63	28.60 (↑5.67)	

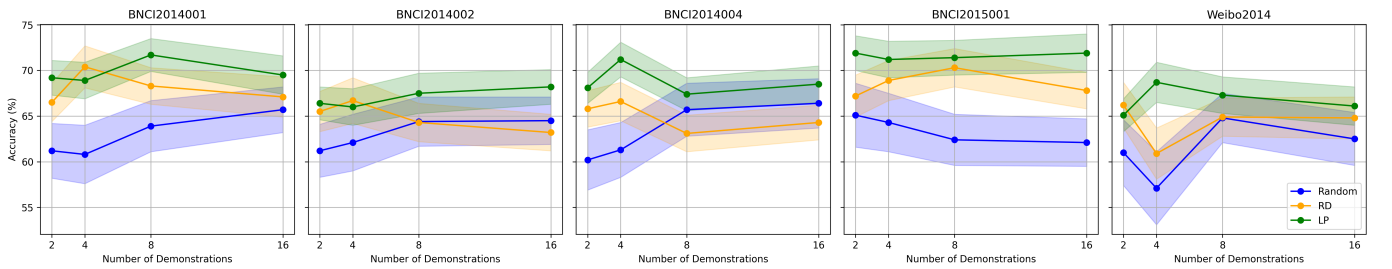


Fig. 1. Performance under different numbers of in-context demonstrations on five MI datasets using BL, RD, and PURE methods. The LLM used here is our base model: Qwen-2.5-14b-instruct-1m.

TABLE VIII
WITHIN-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2015001 USING THE DEEPSEEK-V3 MODEL. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
CTD	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
RD	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
SVM-near	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
SVM-far	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
PURE	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00

In this work, we explored the feasibility of leveraging large language models (LLMs).

This paper proposed a purity-guided unique-round example (PURE) approach for EEG signal classification through in-context using LLMs in a few-shot prompting. The motivation for this study arises from the following key considerations. Although LLMs have demonstrated remarkable in-context learning (ICL) capabilities scenario. Existing demonstration selection methods commonly used in NLP and CV tasks, their application to EEG classification remains largely unexplored. Moreover, as many SOAT LLMs are accessible only through Model-as-a-Service (MaaS) platforms without fine-tuning access, few-shot prompting emerges as a more practical and generalizable approach for leveraging these models. In addition, the scarcity of EEG data compared to text and images severely limits the feasibility of training large domains are often ineffective for EEG data due to its complexity and inherent lack of semantic interpretability. PURE addresses these limitations by integrating domain-specific models, highlighting the importance of bridging this gap by adapting powerful pre-trained LLMs to EEG signal analysis.

To address the unique challenges posed by EEG signals, we incorporated domain-specific feature extraction to obtain compact and discriminative representations, substantially reducing token overhead and enhancing classification performance. Furthermore, we introduced a purity-guided with an active selection strategy that identifies a small set of high-confidence examples with strong intra-class consistency based on local same-label purity. Finally, unlike conventional NLP prompting strategies that select examples for each test instance individually, our method selects a fixed example set once from the training pool, significantly reducing computational cost and inference latency.

, selecting compact sets of representative, low-entropy examples. Extensive experiments across multiple EEG datasets, BCI paradigms, and LLM architectures demonstrated the effectiveness and robustness of the proposed method. The active selection strategy consistently outperformed random selection, delivering superior and BCI paradigms demonstrate that PURE significantly outperforms random selection and other baseline methods, achieving higher accuracy and stability. Moreover, performance improved with the use of more powerful LLMs, and in some cases, surpassed the results of traditional deep neural networks, highlighting the considerable

potential of LLM-based in-context learning approaches for EEG signal analysis and other non-text domains. Furthermore, PURE offers substantial computational efficiency through a one-time selection strategy, highlighting its practical utility in real-world EEG classification applications.

Our future research will extend the proposed method to a broader range of EEG paradigms and classification tasks. We plan to further optimize the prompt structure and incorporate curriculum learning concepts by using demonstrations of varying difficulty levels to explore the impact of example ordering on model performance. These efforts aim to enhance the few-shot prompting capability of LLMs for EEG classification tasks.

Moreover, we envision integrating LLMs with lightweight, plugin-style EEG models. This hybrid architecture would leverage the powerful reasoning and generalization abilities of LLMs while utilizing the precise, parameter-tunable nature of small models trained on high-informativeness samples. Unlike conventional fine-tuning approaches, this design allows for the combination of closed-source LLMs with EEG-specific adaptation without modifying the LLM's parameters, substantially reducing training costs and enhancing practical applicability.

REFERENCES

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, Long Beach, LA, Dec. 2017, pp. 5998–6008.
- [2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, Dec. 2020.
- [3] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *Int'l Conf. on Machine Learning*, Vienna, Austria, Jul. 2021, pp. 8748–8763.
- [4] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds *et al.*, "Flamingo: a visual language model for few-shot learning," *Advances in Neural Information Processing Systems*, vol. 35, pp. 23 716–23 736, 2022.
- [5] Z. Borsos, R. Marinier, D. Vincent, E. Kharitonov, O. Pietquin, M. Sharifi, D. Roblek, O. Teboul, D. Grangier, M. Tagliasacchi *et al.*, "Audiolm: a language modeling approach to audio generation," *IEEE/ACM Trans. on Audio, Speech, and Language Processing*, vol. 31, pp. 2523–2533, 2023.
- [6] H. Zhang, X. Li, and L. Bing, "Video-LLaMa: An instruction-tuned audio-visual language model for video understanding," *arXiv preprint arXiv:2306.02858*, 2023.

- [7] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski *et al.*, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” *arXiv preprint arXiv:2307.15818*, 2023.
- [8] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, “Tabllm: Few-shot classification of tabular data with large language models,” in *Int’l Conf. on Artificial Intelligence and Statistics*, Valencia, Spain, Apr. 2023, pp. 5549–5581.
- [9] X. Liu, D. McDuff, G. Kovacs, I. Galatzer-Levy, J. Sunshine, J. Zhan, M.-Z. Poh, S. Liao, P. Di Achille, and S. Patel, “Large language models are few-shot health learners,” *arXiv preprint arXiv:2305.15525*, 2023.
- [10] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, “Filter bank common spatial pattern (FBCSP) in brain–computer interface,” in *IEEE Int’l Joint Conference on Neural Networks*, Hong Kong, China, Jun. 2008, pp. 2390–2397.
- [11] Q. Dong, L. Li, D. Dai, C. Zheng, J. Ma, R. Li, H. Xia, J. Xu, Z. Wu, B. Chang, X. Sun, L. Li, and Z. Sui, “A survey on in-context learning,” in *Proc. of the 2024 Conf. on Empirical Methods in Natural Language Processing*, Miami, FL, Nov. 2024, pp. 1107–1128.
- [12] J. Liu, D. Shen, Y. Zhang, B. Dolan, L. Carin, and W. Chen, “What makes good in-context examples for GPT-3?” in *Proc. of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, Dublin, Ireland, May. 2022, pp. 100–114.
- [13] J. Li, J. Wang, Z. Zhang, and H. Zhao, “Self-prompting large language models for zero-shot open-domain QA,” in *Proc. of the 2024 Conf. of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Mexico City, Mexico, Jun. 2024, pp. 296–310.
- [14] Y. Zhang, K. Zhou, and Z. Liu, “What makes good examples for visual in-context learning?” vol. 36, New Orleans, LA, Dec. 2023, pp. 17 773–17 794.
- [15] W. Jiang, L. Zhao, and B. liang Lu, “Large brain model for learning generic representations with tremendous EEG data in BCI,” in *The Twelfth Int’l Conf. on Learning Representations*, Vienna, Austria, May. 2024.
- [16] B. Settles, “Active learning literature survey,” University of Wisconsin–Madison, Computer Sciences Technical Report 1648, 2009.
- [17] V. Jayaram and A. Barachant, “Moabb: trustworthy algorithm benchmarking for bcis,” *Journal of Neural Engineering*, vol. 15, no. 6, p. 066011, 2018.
- [18] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz *et al.*, “Review of the BCI competition IV,” *Frontiers in Neuroscience*, vol. 6, p. 55, 2012.
- [19] D. Wu, “Pool-based sequential active learning for regression,” *IEEE Trans. on Neural Networks and Learning Systems*, vol. 30, no. 5, pp. 1348–1359, 2018.
- [20] H. He and D. Wu, “Transfer learning for brain–computer interfaces: A euclidean space data alignment approach,” *IEEE Trans. on Biomedical Engineering*, vol. 67, no. 2, pp. 399–410, 2019.